

The Art Of DESIGN For MANUFACTURE



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Have you ever met with designers who were in stress when their prototypes failed? Or have you seen the production team's anxiety when they were facing manufacturing challenges at production line? Or have you met designers who have had to face wrath of the buyers when their product failed and were questioned for the material used and the product's quality?

In most such cases, it is found that a reliable design for manufacturing (DFM) was not used, by using which problems could have been avoided at the design stage itself. DFM makes a product reliable, sustainable and feasible after production.

Vikas Saxena, general manager - quality at MYBOX Technologies, cites an incident in 2005, when his company almost lost a deal with Ford Motors. During the time, he worked for an electronics manufacturing services (EMS) house in Milwaukee, USA. They were executing an order of Ford's engine control unit boards, which started failing soon after production. They had joint meetings with Ford executives to find out the problem. They realised that since DFM was not used during the design stage, there was a fault in the circuit board where the layers had interchanged, causing a shorting in the boards. The EMS house had to abort the product's development, losing tons of money.

Most electronics giants and auto firms use DFM while their product is in design stage. And DFM is usually done by using certain

tools. The companies have their tools defined as per their own rulebooks. For example, IT giants Cisco and Jabil use Valor, which is an expensive tool but very important for them.

According to experts, DFM should be carried out where the production is to be done, as surface mount technology (SMT) machines differ from company to company. Saxena recalls an instance when a through-hole capacitor started coming out from its pad area during wave soldering, and hence the production had to be stopped immediately. It was found that, since the production centre was in Penang (Malaysia), there was a difference in the SMT machine used and its working, and hence the disaster.

Multinational companies (MNCs) like Jabil, Nokia, Cisco, Apple and GE follow DFM religiously. Jabil uses automatic machines, even for rework, for a perfect result.

Valor: For error-free design

Valor, the most popular DFM tool used by MNCs, provides a PCB manufacturing solution. PCB is the heart of electronics hardware and embedded systems. Valor is a seamless end-to-end 'all-in-one' tool. It helps minimise errors, so that in the long run the designing and manufacturing teams do not have to waste time, labour and cost in fault detection and rectification.

The design, after completion, is checked using Valor or a similar tool. Companies like Cisco have made the use of this DFM tool mandatory as it manufactures up to 24-layer complex printed circuit boards.

Valor has a long checklist that includes boards getting checked on printed circuit boards assembly, layer by layer, including signal check and pattern check on-component basis, count of components, Gerber files, impedance on the PCB, copper layers, copper pads, drill check and through-hole pads. Other than these, quality check of PCBs is done as per application, and thickness, routing, punching, stencil check and stencil paste are also taken into consideration. After the test, Valor differentiates between the hot points and the

When should you start the design for manufacturing process (Image courtesy: <http://svv.io>)

